

Bulk Consumer Data Acquisition and Token Generation Automation for Prepaid Metering System using Python

Reason for the Approach:

We have almost 65k consumers under S&D Chandpur and about 90% of those consumers are using prepaid meters. Every day we get about **100-120** applications and sometimes even more from our consumers for the 'A1' issue which means they need to change the batteries of their meters and they also need tamper tokens to reactivate their meters. So we need to generate those tokens each day. Previously it was done manually which was really time-consuming and at the same time pretty much boring stuff also. Not only that but the token purchase/ recharge history check (Named as Data Reissue in the server) was also done manually which also took long.

What the Approach Does:

In this approach data acquisition for bulk consumers and their token generation are done automatically with the help of python script which completely removes the need of manual labour to accomplish both of the tasks.

How Does the Approach Work:

On receipt of the applications from the consumers, the meter numbers and the mobile numbers of the consumers are entered in an excel file. The python script takes the excel file as an input and reads meter numbers from the file. Then it enters the meter number in the respective field on the webpage of prepaid server and collects the necessary data (ie. last recharge Time/Date, Type etc.) of that particular meter number from the webpage. Finally it stores the necessary data to the respective column of the excel file and saves it in the disk. The process is repeated for all the meter numbers enlisted in that excel file. During token generation process the same approach is followed but this time the collected data includes the generated token and its sequence number. Then the script stores them in the excel file. As the file already contains the mobile numbers, it is now ready to be sent to the particular consumers via API based bulk messaging system.

Extension:

The similar idea has also been deployed for bill download page of postpaid billing server (<http://119.40.95.162:8991/Pages/User/BillPrint.aspx>). This is useful in case of downloading bill copies of bulk consumers who have been shifted to prepaid metering system but their postpaid bill is still due and the **month of postpaid bill closing** varies from consumer to consumer. In this case the **month of last/any postpaid bill** can be collected from “*Bill Information*” page using a python script and then those month data is utilized in “*Customer Bill*” page to download the bill copy of that particular month.